# STACKS, DISCRIMINANTS: LEMMA-DEDEKIND-COMPLEMENTARY-MODULE

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 1694-1699 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 1701-1711, the complete author proof (9 body lines).

*Editorial note.* Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 43 of queue_discriminant.json (pb=555, stmt 1694-1699, proof 1701-1711).

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 0BJF: sources/stacks-discriminant/pinned-original-source.tex, source lines 756–759. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *If the Dedekind different of $A \to B$ is defined, then there is a canonical isomorphism $\mathcal{L}_{B/A} \to \omega_{B/A}$.*

*Proof.* Recall that $\omega_{B/A} = \mathrm{Hom}_A(B, A)$ as $A \to B$ is finite. We send $x \in \mathcal{L}_{B/A}$ to the map $b \mapsto \mathrm{Trace}_{L/K}(bx)$. Conversely, given an $A$-linear map $\varphi : B \to A$ we obtain a $K$-linear map $\varphi_K : L \to K$. Since $K \to L$ is finite étale, we see that the trace pairing is nondegenerate (Lemma 0BJF) and hence there exists a $x \in L$ such that $\varphi_K(y) = \mathrm{Trace}_{L/K}(xy)$ for all $y \in L$. Then $x \in \mathcal{L}_{B/A}$ maps to $\varphi$ in $\omega_{B/A}$. □

*Date*: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.